University of Western Macedonia
Electrical and Computer Engineering Department

/05/2024

Evaluation Questionnaire for UMLcraft

This questionnaire aims to provide answers to certain research questions regarding the effectiveness of the UMLcraft game, which examine the ability to learn, understand, remember and apply the most basic UML diagrams and notations. The process will take place within the framework of the Software Technology course (MK33). Participants will be divided into two groups. Group A (control group) will answer the questionnaire first. Then it will use UMLcraft and it will be reviewed again. Group B will directly use UMLcraft and then answer the questionnaire. *(The data obtained from the questionnaire will remain anonymous.)*

<u>Condescending</u>

1. **I agree to be part of this procedure:** ☐

<u>Personal</u>

2. **What is your gender?**
   ☐ Male
   ☐ Female

<u>Quantitative Assessment</u>

3. **Which of the following diagrams belong to the category of Behavioral Diagrams (Behavioural Diagrams)? (more than one correct choice)**
   α. Activity Diagram
   β. Use Case Diagram
   γ. Class Diagram
   δ. Sequence Diagram

4. **What are the capabilities of Interaction Diagrams? (more than one correct choice)**
   α. are suitable for depicting processes that happen at the same time
   β. showcase cooperation between objects
   γ. are suitable for the detailed description of a behavior
   δ. showcase the behavior of many objects in a use case

5. **Which of the following is true about Activity Diagrams?**
   α. they are used to illustrate parallel processes
   β. they do not provide information about someone performing an action
   γ. they are used to show the workflow of a business activity
   δ. all of the above

-- ^^^ **RQ1**

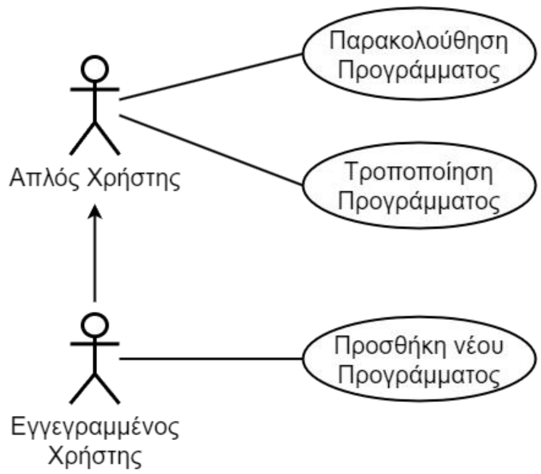
6. **Which of the following is NOT true for Class Diagrams?**
   (more than one correct choice)
   α. in the association relationship between two classes, there is an arrow pointing towards the directed class (target class) but not in the other direction
   β. multiplicities are usually 0, 1, * and do not support values e.g. in the range 2-4
   γ. a public attribute is not accessible from classes in the same package
   δ. cardinalities support any range depending on the requirements

7. **Which of the following apply to an actor (user)?**
   (more than one correct choice)
   α. an actor does not have to be human it can be a subsystem or an external system
   β. an actor is a role that a user plays in relation to the system
   γ. the generalization does not apply to actors
   δ. all of the above

8. **Which of the following apply to a use case?**
   (more than one correct choice)
   α. a use case must be initiated by an actor (user)
   β. a use case records operations that are visible or are not visible to the user
   γ. depicts the time duration of a process
   δ. a use case can have many scenarios

-- ^^^ **RQ2**

9. **From the part of the use case diagram given below, it is concluded that:**

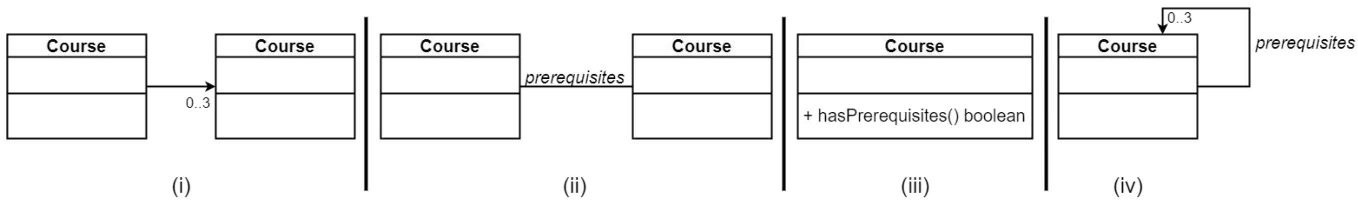


α. a registered user (Εγγεγραμμένος Χρήστης) can call a regular user (Απλός Χρήστης)
β. a regular user (Απλός Χρήστης) is a special category of a registered user (Εγγεγραμμένος Χρήστης)
γ. any user can add a new program
δ. a registered user (Εγγεγραμμένος Χρήστης) can modify a program

10. **Ποια από τις ακόλουθες φράσεις περιγράφει καλύτερα το παρακάτω διάγραμμα κλάσεων;**

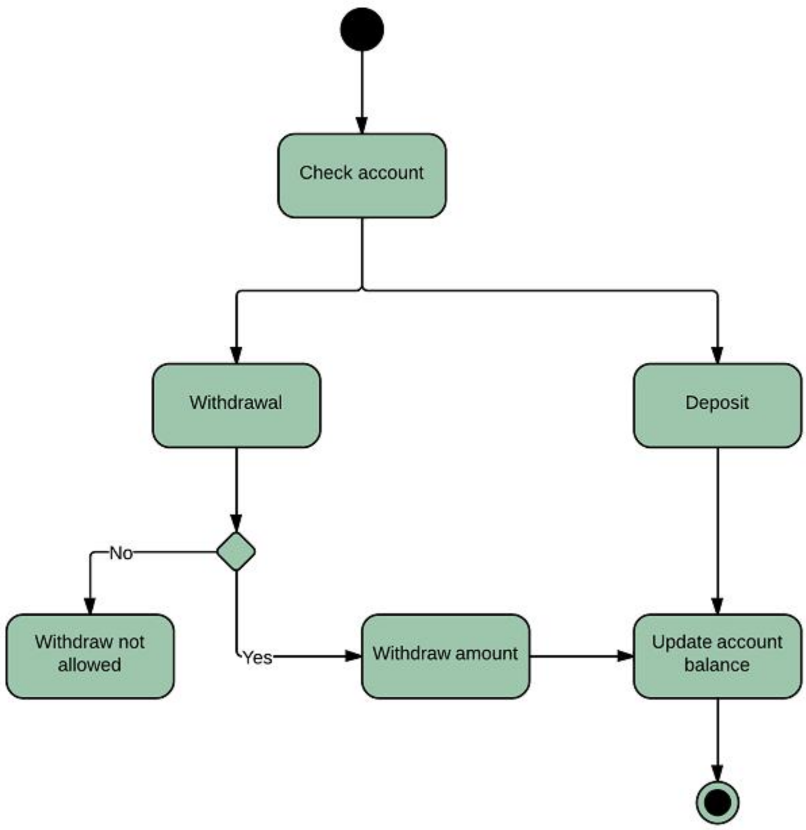
α. a course (Μάθημα) can be attended by 0 to 25 students (Φοιτητές)
β. a student (Φοιτητής) is obliged to attend a course (Μάθημα)
γ. a student (Φοιτητής) is obliged to attend from 0 to 25 courses (Μαθήματα)
δ. a student (Φοιτητής) attends from 0 to25 courses (Μαθήματα)

11. **A course at a University can have up to 3 prerequisite courses.**
Which of the UML class diagrams below best represents this statement?



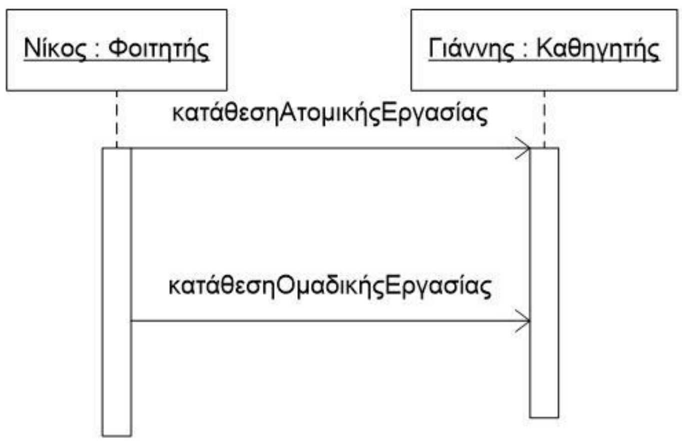
α. (iii)
β. (iv)
γ. (i)
δ. (ii)

12. **What type of diagram is the following and what function does it perform?**



α. Activity Diagram, possible banking transactions
β. Sequence Diagram, commercial transaction
γ. Activity Diagram, commercial transaction
δ. Class Diagram, banking transaction

13. **From the following part of a Sequence Diagram we conclude that:**



α. the classes student (Φοιτητής) and professor (Καθηγητής) have some kind of correlation between them
β. the submission of GroupWork (κατάθεσηΟμαδικήςΕργασίας) is done in parallel with the submission of IndividualWork (κατάθεσηΑτομικήςΕργασίας)
γ. John (Γιάννης) is an object of the class student (Φοιτητής)
δ. the class student (Φοιτητής) has a function "submitIndividualWork" (κατάθεσηΑτομικήςΕργασίας)

--- ^^^ **RQ3**

14. **Draw the class diagram for the following problem.**

  «A personnel access control system has been installed at the entrance of a company. Company executives enter with fingerprint recognition. Company technicians working in external facilities have a card with a unique code. Each card corresponds to a technician and each technician has only one such card. In order for an employee to enter or exit the building, their identity must be validated using a card or fingerprint, which also function as keys to the building's entrance door managed by the control system. The system keeps a table with the times when an entry or exit (movement) took place for each employee.»

-- ^^^ **RQ4**

Qualitative Assessment (After using UMLcraft)

15. **Did UMLcraft help you learn the basic principles of UML more easily in relation to its usage and the basic types of diagrams it includes?**

	1	2	3	4	5	
Not at all	◯	◯	◯	◯	◯	Extremely

16. **Did UMLcraft help you remember the basic principles of UML more easily in relation to the rules and symbols of the diagrams it includes?**

	1	2	3	4	5	
Not at all	◯	◯	◯	◯	◯	Extremely

17. **Did UMLcraft help you understand and describe the function of given diagrams more easily?**

	1	2	3	4	5	
Not at all	◯	◯	◯	◯	◯	Extremely

18. **Did UMLcraft help you apply the basic principles of UML in the context of implementing a diagram more easily?**

	1	2	3	4	5	
Not at all	◯	◯	◯	◯	◯	Extremely

19. **What did you like about UMLcraft, did it help you improve in relation to UML and if not, what did you not understand about it;**

-- ^^^ **RQ5**

Fill in the tables after solving each exercise.

Activity Diagram

	Minigame 1	**Minigame 2**	**Exercise 1**	**Exercise 2**	**Exercise 3**
Time (m:s)					
Gems					

Use Case Diagram

	Minigame 1	**Exercise 1**	**Exercise 2**	**Exercise 3**
Time (m:s)				
Gems				

Sequence Diagram

	Minigame 1	**Minigame 2**	**Exercise 1**	**Exercise 2**	**Exercise 3**
Time (m:s)					
Gems					

Class Diagram

	Minigame 1	**Minigame 2**	**Minigame 3**	**Exercise 1**	**Exercise 2**	**Exercise 3**
Time (m:s)						
Gems						